



Computer Networks Week # 11

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What is Subnet Mask?

IPv4 address has two components, the network part and the host part.

In fact **IPv4 address** , is a combination of IPv4 address and Subnet mask

The purpose of subnet mask is to identify which part is the **network part** and which part is the **host part**.

Subnet mask is a 32 bit number where all the bits of the network part are represented as **1** and all the bits of the host part are represented as **0**.

For example, for a **Class C Network**, 192.168.10.0, the address part and the subnet mask is **255.255.255.0**

Subnet and Subnetting

- A logical, visible subdivision of an IP network is called **subnet** or **subnetwork**.
 - It is created by dividing the host identifier
- **Subnetting** is the practice of dividing a network into two or more networks.
- **In subnetting**, a class A or class B or class C block is divided into several subnets (each subnet with larger prefix length than the original network).
- **For example**, divide the **class A** into **four subnets**, then each subnet will have prefix length as 10 (take two bits from host id part in order to obtain subnets).

Subnetting

Subnetting is done by taking the bits from host part and adding it to the network part

Remember the following [as already covered]

If all the bits in the host part are "0", that represents the **network id** (network address)

If all the bits in the host part are "0" except the last bit (R_{Mb}), it is the **first usable IPv4 address**

If all the bits in the host part are "1" except the last bit (R_{Mb}), it is the **last usable IPv4 address**

If all the bits in the host part are "1", that represents the **broadcast**

Class C – 1 bit subnetting

Consider class c network 192.168.10.0 (subnet mask is 255.255.255.0)

- ❖ If we include one bit from the host part to the network part, the **subnet mask** changes into 255.255.255.128.
- ❖ The single bit can have two values in last octet, either 0 or 1 (so we can get **two subnets with a single bit subnetting**)
- ❖ 11000000.10101000.00001010.**0** | 00000000
11111111.11111111.11111111.**1** | 00000000
- ❖ So the network 192.168.10.0 is divided into two networks,
 - ❖ each network has 128 total addresses of which 126 are usable
 - ❖ two are used in each subnet to represent the network address and broadcast address.
 - ❖ The subnet mask for one bit subnetting is **255.255.255.128**.

Class C – 1 bit subnetting

SN No.	Description	Binaries	Decimal
1	Network Address	11000000.10101000.00001010.00000000	192.168.10.0
	First usable address	11000000.10101000.00001010.00000001	192.168.10.1
	Last usable address	11000000.10101000.00001010.01111110	192.168.10.126
	Broadcast Address	11000000.10101000.00001010.01111111	192.168.10.127
2	Network Address	11000000.10101000.00001010.10000000	192.168.10.128
	First usable address	11000000.10101000.00001010.10000001	192.168.10.129
	Last usable address	11000000.10101000.00001010.11111110	192.168.10.254
	Broadcast Address	11000000.10101000.00001010.11111111	192.168.10.255

Class C – 2 bit subnetting

SN No.	Description	Binaries	Decimal
1	NA	11000000.10101000.00001010. 00 000000	192.168.10.0
	1st	11000000.10101000.00001010. 00 000001	192.168.10.1
	Last	11000000.10101000.00001010. 00 111110	192.168.10.62
	BA	11000000.10101000.00001010. 00 111111	192.168.10.63
2	NA	11000000.10101000.00001010. 01 000000	192.168.10.64
	1st	11000000.10101000.00001010. 01 000001	192.168.10.65
	Last	11000000.10101000.00001010. 01 111110	192.168.10.126
	BA	11000000.10101000.00001010. 01 111111	192.168.10.127
3	NA	11000000.10101000.00001010. 10 000000	192.168.10.128
	1st	11000000.10101000.00001010. 10 000001	192.168.10.129
	Last	11000000.10101000.00001010. 10 111110	192.168.10.190
	BA	11000000.10101000.00001010. 10 111111	192.168.10.191
4	NA	11000000.10101000.00001010. 11 000000	192.168.10.192
	1st	11000000.10101000.00001010. 11 000001	192.168.10.193
	Last	11000000.10101000.00001010. 11 111110	192.168.10.254
	BA	11000000.10101000.00001010. 11 111111	192.168.10.255

Class C – 3 bit subnetting

SN No.	Description	Binaries	Decimal
1	NA	11000000.10101000.00001010. 000 00000	192.168.10.0
	1st	11000000.10101000.00001010.00000001	192.168.10.1
	Last	11000000.10101000.00001010.00011110	192.168.10.30
	BA	11000000.10101000.00001010. 000 11111	192.168.10.31
2	NA	11000000.10101000.00001010. 001 00000	192.168.10.32
	1st	11000000.10101000.00001010.00100001	192.168.10.33
	Last	11000000.10101000.00001010.00111110	192.168.10.62
	BA	11000000.10101000.00001010. 001 11111	192.168.10.63
.....			
8	NA	11000000.10101000.00001010. 111 00000	192.168.10.224
	1st	11000000.10101000.00001010.11100001	192.168.10.225
	Last	11000000.10101000.00001010.11111110	192.168.10.254
	BA	11000000.10101000.00001010. 111 11111	192.168.10.255

Class B – 1 bit subnetting

Consider class B network 172.16.0.0 (subnet mask is 255.255.0.0)

- ❖ If we include one bit from the host part to the network part, the **subnet mask** changes into 255.255.128.0.
- ❖ The single bit can have two values in last octet, either 0 or 1 (so we can get **two subnets with a single bit subnetting**)
- ❖ 10101100.00010000.**0** | 00000000.00000000
11111111.11111111.**1** | 00000000.00000000
- ❖ So the network 172.16.0.0 is divided into two networks,
 - ❖ each network has 32768 total addresses of which 32766 are usable
 - ❖ two are used in each subnet to represent the network address and broadcast address.
 - ❖ The subnet mask for one bit subnetting is **255.255.128.0**.

Class B – 1 bit subnetting

SN No.	Description	Binaries	Decimal
1	Network Address	10101100.00010000. 0 0000000.00000000	172.16.0.0
	First address	10101100.00010000. 0 0000000.00000001	172.16.0.1
	Last address	10101100.00010000. 0 1111111.11111110	172.16.127.254
	Broadcast Address	10101100.00010000. 0 1111111.11111111	172.16.127.255
2	Network Address	10101100.00010000. 1 0000000.00000000	172.16.128.0
	First address	10101100.00010000. 1 0000000.00000001	172.16.128.1
	Last address	10101100.00010000. 1 1111111.11111110	172.16.255.254
	Broadcast Address	10101100.00010000. 1 1111111.11111111	172.16.255.255

Class B – 2 bit subnetting

SN No.	Description	Binaries	Decimal
1	NA	10101100.00010000. 00 000000.00000000	172.16.0.0
	1st	10101100.00010000. 00 000000.00000001	172.16.0.1
	Last	10101100.00010000. 00 111111.11111110	172.16.63.254
	BA	10101100.00010000. 00 111111.11111111	172.16.63.255
2	NA	10101100.00010000. 01 000000.00000000	172.16.64.0
	1st	10101100.00010000. 01 000000.00000001	172.16.64.1
	Last	10101100.00010000. 01 111111.11111110	172.16.127.254
	BA	10101100.00010000. 01 111111.11111111	172.16.127.255
3	NA	10101100.00010000. 10 000000.00000000	172.16.128.0
	1st	10101100.00010000. 10 000000.00000001	172.16.128.1
	Last	10101100.00010000. 10 111111.11111110	172.16.191.254
	BA	10101100.00010000. 10 111111.11111111	172.16.191.255
4	NA	10101100.00010000. 11 000000.00000000	172.16.192.0
	1st	10101100.00010000. 11 000000.00000001	172.16.192.1
	Last	10101100.00010000. 11 111111.11111110	172.16.192.254
	BA	10101100.00010000. 11 111111.11111111	172.16.192.255

Class A – 1 bit subnetting

Consider class A network 10.0.0.0 (subnet mask is 255.0.0.0)

- ❖ If we include one bit from the host part to the network part, the **subnet mask** changes into 255.128.0.0.
- ❖ The single bit can have two values in second octet, either 0 or 1 (so we can get **two subnets with a single bit subnetting**)
- ❖ 00001010.**0** | 00000000.00000000.00000000
11111111.**1** | 00000000.00000000.00000000
- ❖ The network 10.0.0.0 is divided into two networks, each network has 8388608 total IPv4 Addresses and 8388606 usable IPv4 Addresses (two IPv4 Addresses are used in each subnet to represent the network address and the directed broadcast address).

Class A – 1 bit subnetting

SN No	Description	Binaries	Decimal
1	Network Address	00001010. 0 0000000.00000000.00000000	10.0.0.0
	First IPv4 address	00001010.00000000.00000000.00000001	10.0.0.1
	Last IPv4 address	00001010.01111111.11111111.11111110	10.127.255.254
	Broadcast Address	100001010. 0 1111111.11111111.11111111	10.127.255.255
2	Network Address	00001010. 1 0000000.00000000.00000000	10.128.0.0
	First IPv4 address	00001010.10000000.00000000.00000001	10.128.0.1
	Last IPv4 address	00001010.11111111.11111111.11111110	10.255.255.254
	Broadcast Address	00001010. 1 1111111.11111111.11111111	10.255.255.255

Class A – 2 bit subnetting

N No	Description	Binaries	Decimal
1	Network Address	00001010. 00 000000.00000000.00000000	10.0.0.0
	First IPv4 address	00001010.00000000.00000000.00000001	10.0.0.1
	Last IPv4 address	00001010.00111111.11111111.11111110	10.63.255.254
	Broadcast Address	00001010. 00 111111.11111111.11111111	10.63.255.255
2	Network Address	00001010. 01 000000.00000000.00000000	10.64.0.0
	First IPv4 address	00001010.01000000.00000000.00000001	10.64.0.1
	Last IPv4 address	00001010.01111111.11111111.11111110	10.127.255.254
	Broadcast Address	00001010. 01 111111.11111111.11111110	10.127.255.255
3	Network Address	00001010. 10 000000.00000000.00000000	10.128.0.0
	First IPv4 address	00001010.10000000.00000000.00000001	10.128.0.1
	Last IPv4 address	00001010.10111111.11111111.11111110	10.191.255.254
	Broadcast Address	00001010. 10 111111.11111111.11111111	10.191.255.255
4	Network Address	00001010. 11 000000.00000000.00000000	10.192.0.0
	First IPv4 address	00001010.11000000.00000000.00000001	10.192.0.1
	Last IPv4 address	00001010.11111111.11111111.11111110	10.255.255.254
	Broadcast Address	00001010. 11 111111.11111111.11111111	10.255.255.255

Example: Route 193.205.102.36

193								205								102								36							
1	1	0	0	0	0	0	1	1	1	0	0	1	1	0	1	0	1	1	0	0	1	1	0	0	0	1	0	0	1	0	0

Class C address;

Outside private domain routed with mask 255.255.255.0

network																				host											
193								205								102								36							
1	1	0	0	0	0	0	1	1	1	0	0	1	1	0	1	0	1	1	0	0	1	1	0	0	0	1	0	0	1	0	0

Inside private domain, administrator has set netmask 255.255.255.248

255								255								255								248							
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0

Hence, route to subnet address and then to host id, computed as:

network																				subnet					host				
1	1	0	0	0	0	0	1	1	1	0	0	1	1	0	1	0	1	1	0	0	1	1	0	0	0	0	1	0	0
193.205.102.32 /29																							4						

Classless Addressing

- Classless addressing is a short term solution to solve the address depletion problem
 - Uses the same address space but change the distribution of addresses to provide a fair share to each organization.
 - Still uses the IPv4 addresses (class privilege was removed from the distribution)
- The long-range solution already devised called **IPv6**
 - The larger address space is obtained by **increasing the length of IP addresses** (128 bits)
 - It means that **format of IP** packets need to be changed

Classless Addressing

- In classless addressing, **variable-length blocks** are assigned that belong to no class.
 - the entire address space is divided into **blocks of different sizes**.
- **Classless addressing**
 - uses a variable number of bits for the network and host portions of the address.
 - treats the IP address as a 32 bit stream of ones and zeroes,
 - where the boundary between network and host portions can fall anywhere between bit 0 and bit 31.

Classless Addressing

- How to find the prefix length if an address is given?
- As prefix length is not inherent in the address
 - Need to separately give the length of the prefix
- So, Prefix length is added to the address, separated by a slash
- The notation is informally referred to as slash notation and formally as *classless interdomain routing or CIDR*

Format of classless addressing address



Classless Addressing

- The network portion of an IP address is determined by how many 1's are in the subnet mask.
- **A Subnet Mask is used to divide the IP address into network and host addresses**
- **A mask is a 32-bit number in which the n leftmost bits are 1s and the $32 - n$ rightmost bits are 0s**
- In $x.y.z.t/n$
 - $x.y.z.t$ defines one of the addresses and
 - the $/n$ defines the mask

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Table Prefix lengths

<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>
/1	128.0.0.0	/9	255.128.0.0	/17	255.255.128.0	/25	255.255.255.128
/2	192.0.0.0	/10	255.192.0.0	/18	255.255.192.0	/26	255.255.255.192
/3	224.0.0.0	/11	255.224.0.0	/19	255.255.224.0	/27	255.255.255.224
/4	240.0.0.0	/12	255.240.0.0	/20	255.255.240.0	/28	255.255.255.240
/5	248.0.0.0	/13	255.248.0.0	/21	255.255.248.0	/29	255.255.255.248
/6	252.0.0.0	/14	255.252.0.0	/22	255.255.252.0	/30	255.255.255.252
/7	254.0.0.0	/15	255.254.0.0	/23	255.255.254.0	/31	255.255.255.254
/8	255.0.0.0	/16	255.255.0.0	/24	255.255.255.0	/32	255.255.255.255

Thus, Classful addressing is a special case of classless addressing

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

Example 1: What is the first address in the block if one of the addresses is **167.199.170.82/27**?

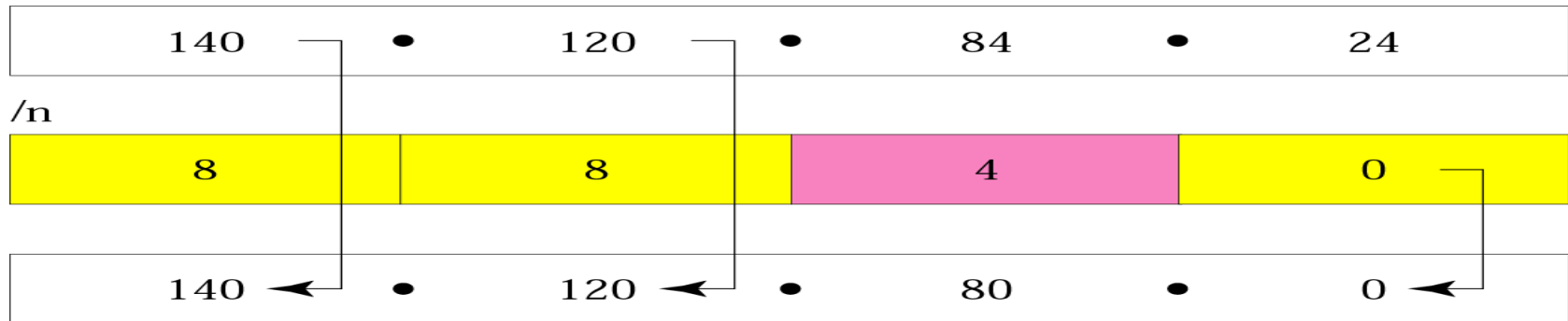
Solution: The prefix length is 27, which means that we must keep the first 27 bits as is and change the remaining bits (5) to 0s. The following shows the process:

```
Address in binary:      10100111 11000111 10101010 01010010
Keep the left 27 bits:  10100111 11000111 10101010 01000000
Result in CIDR notation: 167.199.170.64/27
```

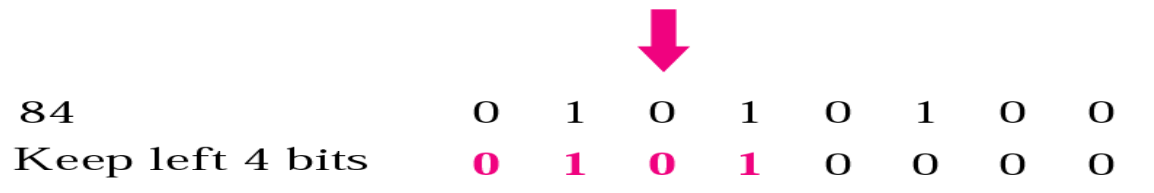
Example 2: What is the first address in the block if one of the addresses is **140.120.84.24/20**?

Answer: The first address is 140.120.80.0/20

IP Address



First Address



Result in decimal: 80

Write 84 as sum of:

128	64	32	16	8	4	2	1
0	64	0	16	0	4	0	0

Select only leftmost 4:

0	64	0	16
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Add to find the result: 80

Example 3: Find the number of addresses in the block if one of the addresses is **140.120.84.24/20**.

Solution: The prefix length is 20. The number of addresses in the block is 2^{32-20} or 2^{12} or **4096**. Note that this is a large block with **4096** addresses.

Note: The number of addresses can also be found by complementing the mask, interpreting it as a decimal number, and adding 1 to it.

Example 4: find the first and last address in the block if one of the addresses is **140.120.84.24/20**.

Answer: The **first** address is **140.120.80.0/20** (set all bits of host part to 0)

The **last** address is **140.120.95.255/20** (set all bits of host part to 1)

Another way to find the last address:

Add the **mask complement** to the **beginning address** to find the **last address**.

140 . 120 . 80 . 0

0 . 0 . 15 . 255

140 . 120 . 95 . 255

Example 5: Find the block if one of the addresses is 190.87.140.202/29.

Solution: To find the **first** address, we notice that the mask (/29) has five 1s in the last byte. So write the last byte as powers of 2 and retain only the leftmost five as shown below:

202 $\Rightarrow 128 + 64 + 0 + 0 + 8 + 0 + 2 + 0$

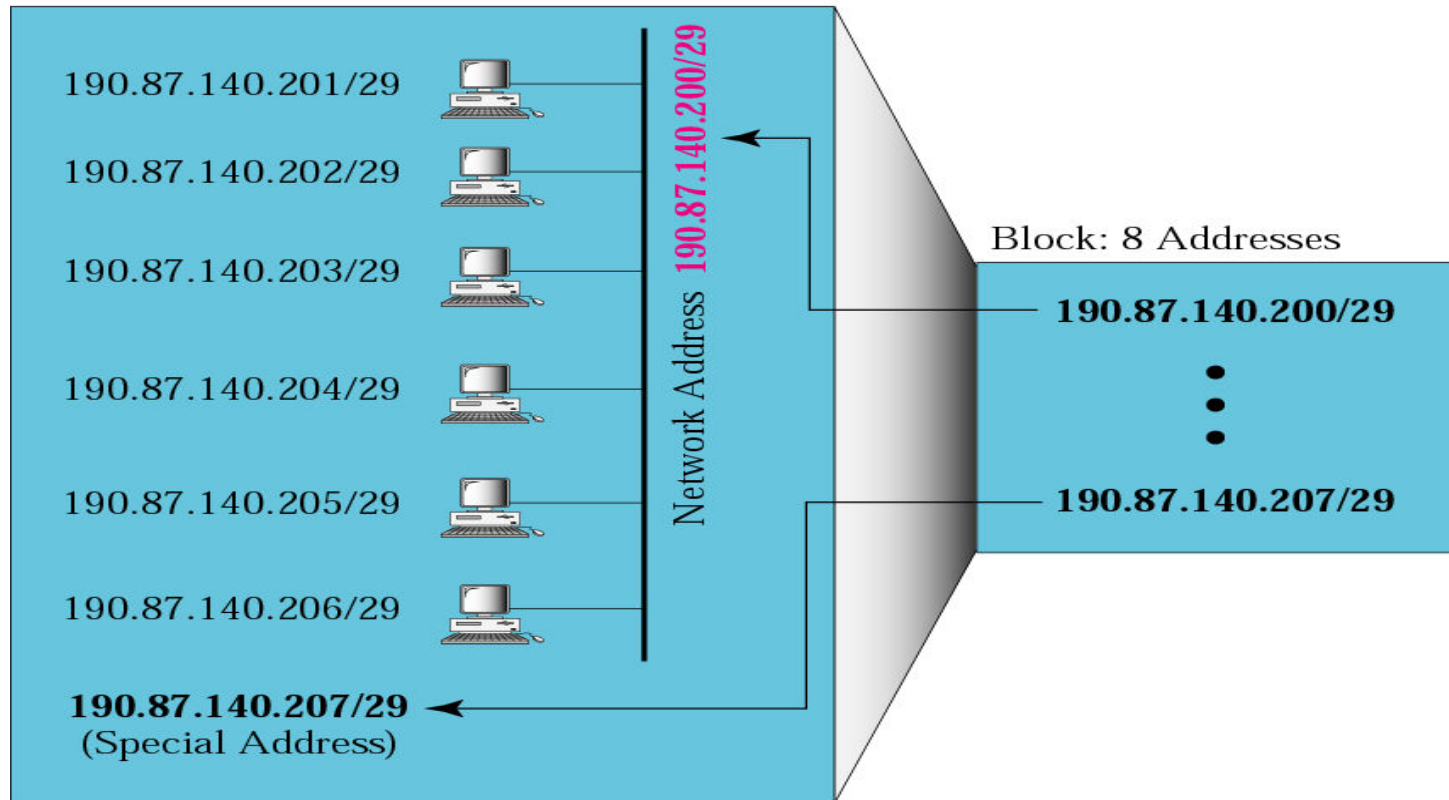
The leftmost 5 numbers are $\Rightarrow 128 + 64 + 0 + 0 + 8$

The first address is 190.87.140.200/29

The number of addresses is 2^{32-29} or 8. To find the last address, use the complement of the mask. The mask has twenty-nine 1s; the complement has three 1s. The complement is 0.0.0.7. Add this to the first address to get **190.87.140.207/29**. So, the first address is **190.87.140.200/29**, the last address is **190.87.140.207/29**. There are only 8 addresses in this block.

Previous Example: Network Configuration

Network Organization



In classless addressing, the last address in the block does not necessarily end in 255.

➤ In classless addressing, an address can belong to many blocks (depending on value of prefix associated with that block).

➤ For example, the address **230.8.24.56** can belong to

Prefix length	Block	
	From	To
16	230.8.0.0	230.8.255.255
20	230.8.16.0	230.8.31.255
26	230.8.24.0	230.8.24.63
27	230.8.24.32	230.8.24.63
29	230.8.24.56	230.8.24.63
31	230.8.24.56	230.8.24.57

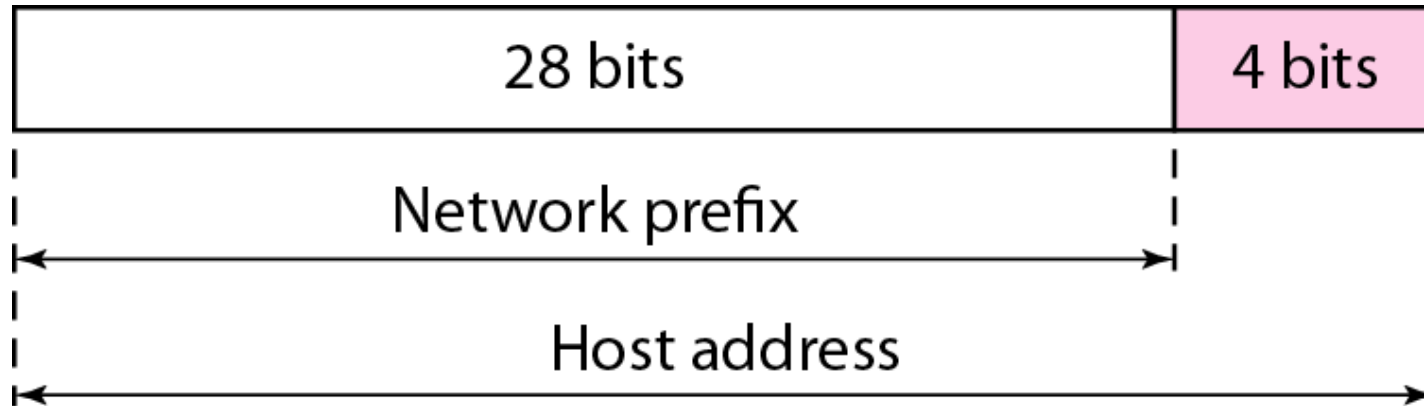
Variable Length Subnet Mask (VLSM)

- VLSM is a way of further subnetting a subnet.
- we can allocate IPv4 addresses to the subnets by the exact need by using VLSM
- VLSM allows us to use more than one subnet mask within the same network address space.
- In classful addressing, we can divide a network only into subnets with equal number of IPv4 addresses.
- VLSM allows to create subnets from a single network with unequal number of IPV4 addresses

Variable Length Subnet Mask (VLSM)

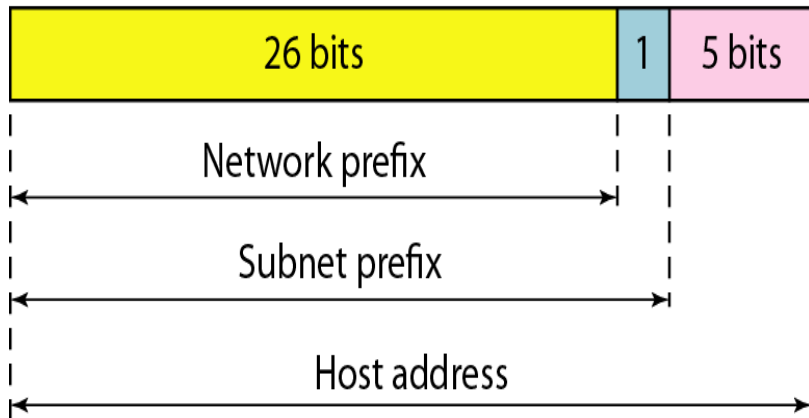
- Suppose we want to divide 192.168.10.0 (a Class C network) into four networks each with unequal number of address requirements as given.
 - Subnet A : 126 IPv4 Addresses.
 - Subnet B : 62 IPv4 Addresses.
 - Subnet C : 30 IPv4 Addresses.
 - Subnet D : 30 IPv4 Addresses.
- Such division is not possible in classful addressing, since it divide the network equally,
- **but is possible with VLSM.**

Two-level hierarchy in an IPv4 address

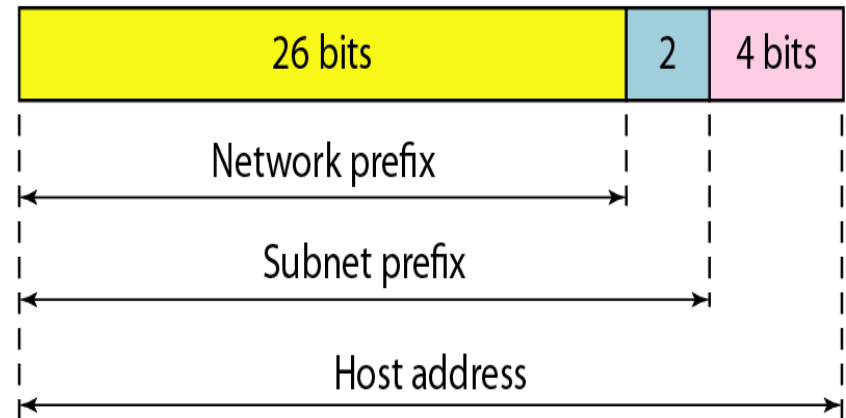


Three-level hierarchy in an IPv4 address

Subnet 1



Subnets 2 and 3



Variable Length Subnet Mask (VLSM)

Division of **192.168.10.0/24** (original network) into four networks with VLSM.

FIRST DIVISION

- Divide into two networks equally with 128 addresses (126 usable) using subnet mask 255.255.255.128
- Two subnets each with 128 addresses
 1. 192.168.10.0/25 [255.255.255.128]
 - 11000000.10101000.00001010.00000000 [in binary]
11111111.11111111.11111111.10000000 [subnet mask]
 2. 192.168.10.128/25 [255.255.255.128]
 - 11000000.10101000.00001010.10000000
11111111.11111111.11111111.10000000 [subnet mask]

Variable Length Subnet Mask (VLSM)

SECOND DIVISION

- Divide second subnet 192.168.10.128/25 (obtained from first division) again into two networks, each with 64 addresses (62 usable) using subnet mask 255.255.255.192
- Two subnets each with 64 addresses
 1. 192.168.10.128/26 [255.255.255.192]
 - 11000000.10101000.00001010.10000000 [in binary]
 - 11111111.11111111.11111111.11000000 [subnet mask]
 2. 192.168.10.192/26 [255.255.255.192]
 - 11000000.10101000.00001010.10000000 [in binary]
 - 11111111.11111111.11111111.11000000 [subnet mask]

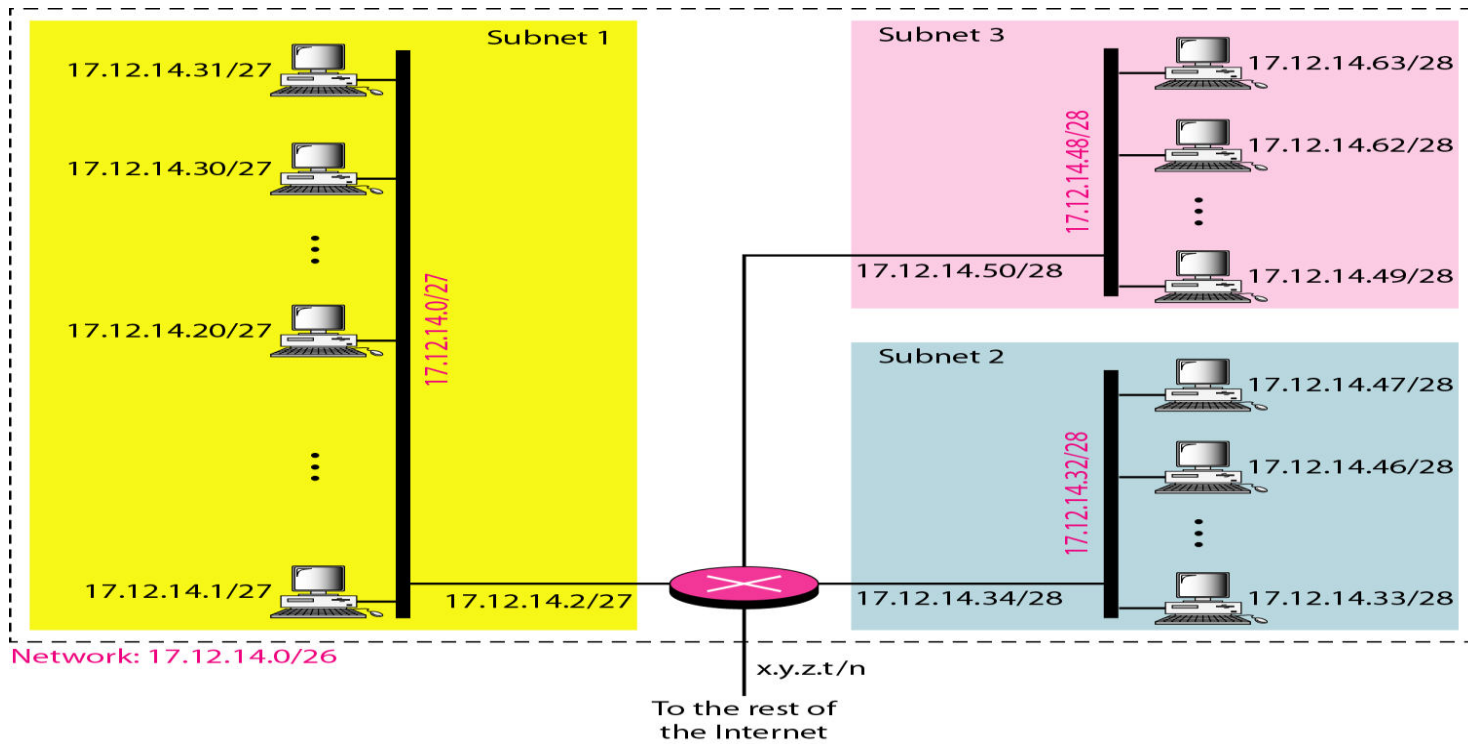
Variable Length Subnet Mask (VLSM)

THIRD DIVISION

- Divide second subnet 192.168.10.192/26 (obtained from second division) again into two networks, each with 32 addresses (30 usable) using subnet mask 255.255.255.224
- Two subnets each with 64 addresses
 1. 192.168.10.192/27 [255.255.255.224]
 - 11000000.10101000.00001010.11000000 [in binary]
11111111.11111111.11111111.11100000 [subnet mask]
 2. 192.168.10.224/27 [255.255.255.224]
 - 11000000.10101000.00001010.11100000
11111111.11111111.11111111.11100000 [subnet mask]
- So, splitting of 192.168.10.0/24 into four subnets using VLSM with unequal number of addresses is done

Example: An organization is granted a block of addresses starting with 17.12.14.0/26 (64 addresses). The organization needs to have three sub-blocks of addresses to use in its three subnets: one sub-block of 32 addresses, and two sub-blocks of 16 addresses each. Design the sub-blocks and find out how many addresses are still available after these allocations.

Configuration and addresses in a subnetted network



Example: An organization is granted a block of addresses starting with 14.24.74.0/24. The organization needs to have three sub-blocks of addresses to use in its three subnets: one sub-block of 10 addresses, one sub-block of 60 addresses, and one sub-block of 120 addresses.

Design the sub-blocks and find out how many addresses are still available after these allocations.

Solution: There are $2^{32-24} = 256$ addresses in this block. The first address is 14.24.74.0/24 and last address is 14.24.74.255/24 .

Assign addresses to subblocks starting with the largest and ending with the smallest one.

➤ Mask n_1 for the **first** (largest) subnet 2^{32-n_1} must be 128 (a number with power of 2 nearest to 120). **So $n_1 = 25$** . We allocate 128 addresses instead of 120 to this subnet

➤ The first address in this subnet is 14.24.74.0/25 and last address is 14.24.74.127/25

Solution (previous example continue)

➤ Mask for the **second subnet** 2^{32-n_2} must be 64 (a number with power of 2 nearest to 60). So **$n_2 = 26$**

➤ The first address in this subnet is 14.24.74.128/26 and last address is 14.24.74.191/26

➤ Mask for the **third subnet** 2^{32-n_3} must be 16 (a number with power of 2 nearest to 10). So **$n_3 = 28$**

➤ The first address in this subnet is 14.24.74.192/28 and last address is 14.24.74.207/28

➤ So we have $128+64+16 = 208$ addresses in all three subblocks.

➤ Therefore, 48 address are still left in reserve.

Example

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- a. The first group has 64 customers; each needs 256 addresses.
- b. The second group has 128 customers; each needs 128 addresses.
- c. The third group has 128 customers; each needs 64 addresses.

Design the subblocks and find out how many addresses are still available after these allocations.

Example (continued)

Solution

Group 1

For this group, each customer needs 256 addresses. This means that 8 ($\log_2 256$) bits are needed to define each host. The prefix length is then $32 - 8 = 24$. The addresses are

<i>1st Customer:</i>	<i>190.100.0.0/24</i>	<i>190.100.0.255/24</i>
<i>2nd Customer:</i>	<i>190.100.1.0/24</i>	<i>190.100.1.255/24</i>
<i>...</i>		
<i>64th Customer:</i>	<i>190.100.63.0/24</i>	<i>190.100.63.255/24</i>
<i>Total = $64 \times 256 = 16,384$</i>		

Example (continued)

Group 2

For this group, each customer needs 128 addresses. This means that 7 ($\log_2 128$) bits are needed to define each host. The prefix length is then $32 - 7 = 25$. The addresses are

<i>1st Customer:</i>	<i>190.100.64.0/25</i>	<i>190.100.64.127/25</i>
<i>2nd Customer:</i>	<i>190.100.64.128/25</i>	<i>190.100.64.255/25</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.127.128/25</i>	<i>190.100.127.255/25</i>
<i>Total = $128 \times 128 = 16,384$</i>		

Example (continued)

Group 3

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host. The prefix length is then $32 - 6 = 26$. The addresses are

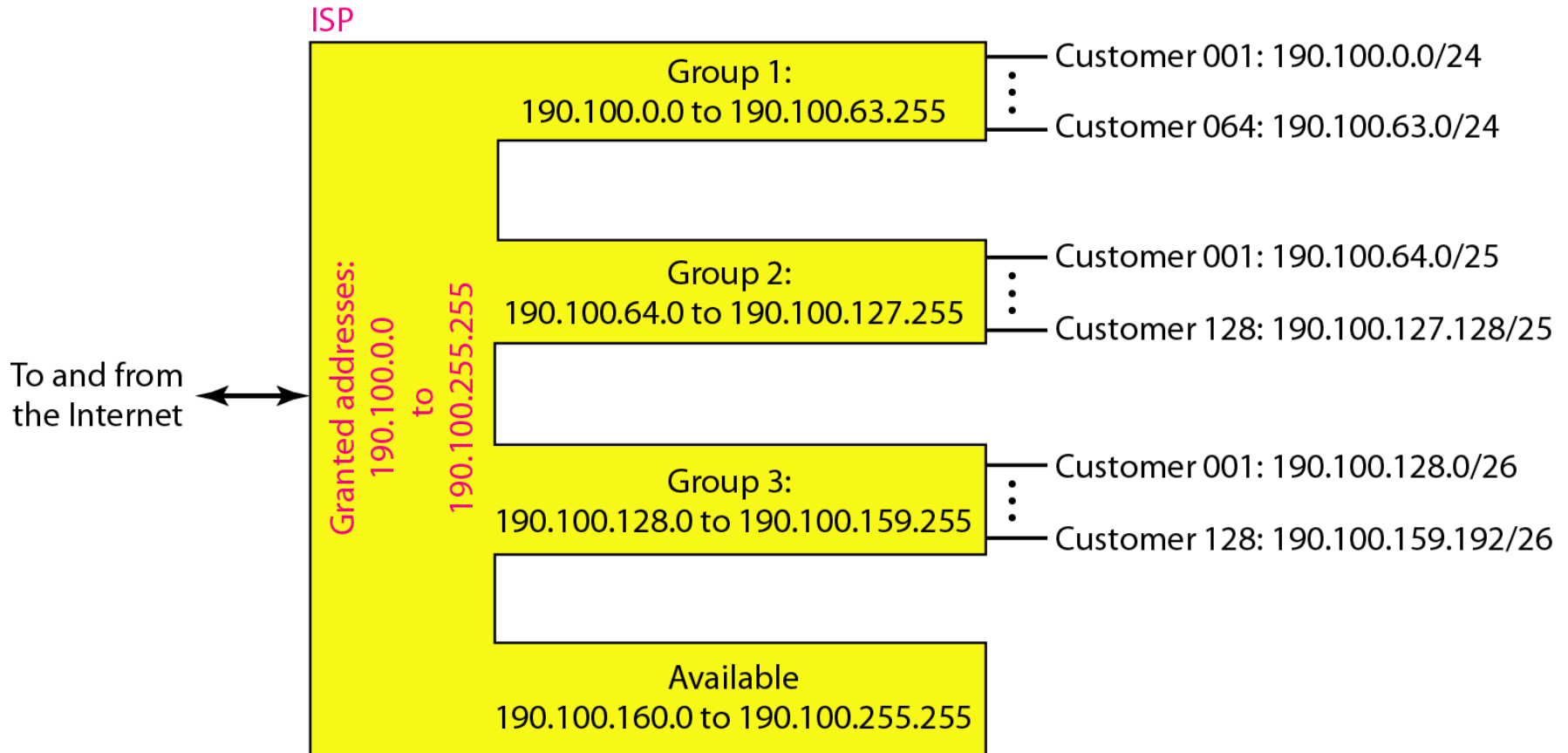
1st Customer:	190.100.128.0/26	190.100.128.63/26
2nd Customer:	190.100.128.64/26	190.100.128.127/26
...		
128th Customer:	190.100.159.192/26	190.100.159.255/26
Total =	$128 \times 64 = 8192$	

Number of granted addresses to the ISP: 65,536

Number of allocated addresses by the ISP: 40,960

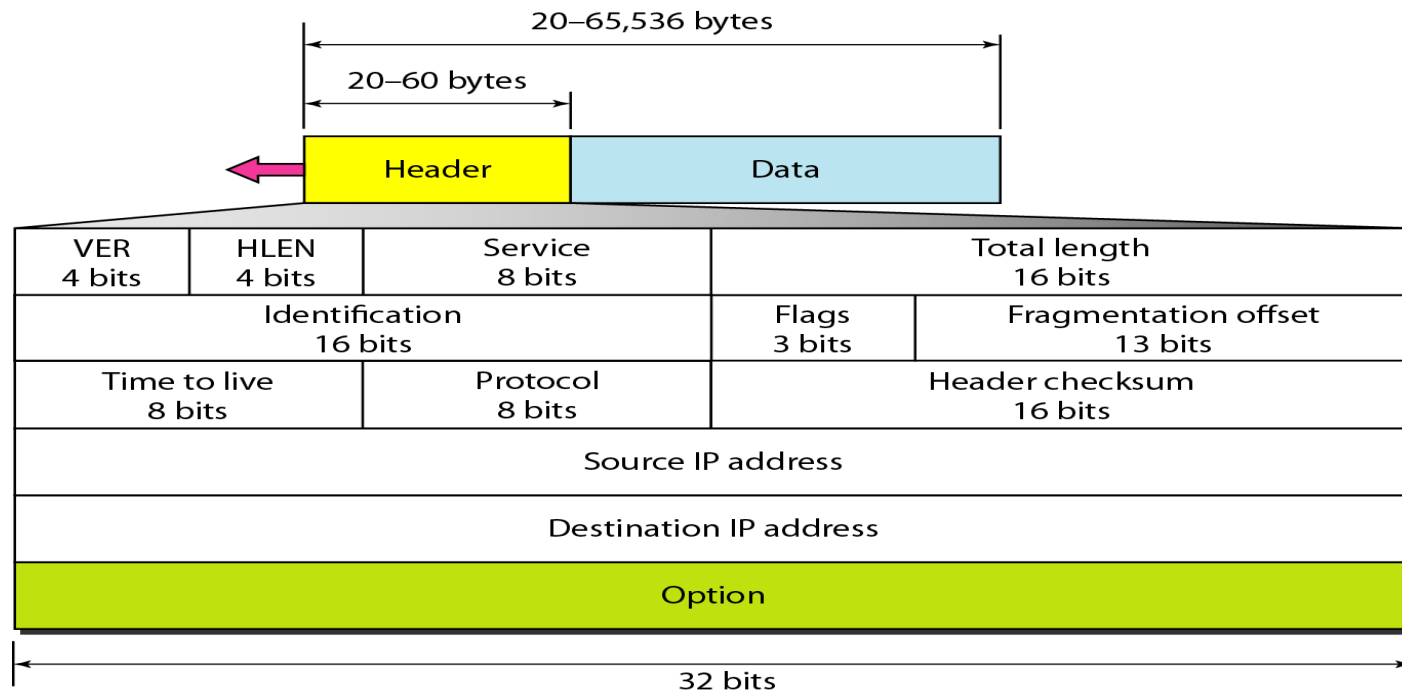
Number of available addresses: 24,576

Figure: An example of address allocation and distribution by an ISP



IPv4 datagram format

- A datagram is a variable-length packet consisting of two parts: header and data.
- The header is 20 to 60 bytes in length
 - contains information essential to routing and delivery



IPv4 datagram format

Header length (HLEN):

- 4-bit field defines the total length of the datagram header in 4-byte words.
- When there are no options, the header length is 20 bytes, and the value of this field is 5 ($5 \times 4 = 20$).
- When the option field is at its maximum size, the value of this field is 15 ($15 \times 4 = 60$).

IPv4 datagram format

Total length:

- A 16-bit field that defines the total length (header plus data) in bytes (limited to 65,535 ($2^{16} - 1$) bytes).

Length of data = total length - header length

- some physical networks are not able to encapsulate a datagram of 65,535 bytes in their frames
 - datagram must be fragmented to be able to pass through those networks

Example: An IPv4 packet has arrived with the first 8 bits as shown: **01000010**. The receiver discards the packet. Why?

Solution: There is an error in this packet.

The 4 leftmost bits (0100) show the version, which is correct.

The next 4 bits (0010) show an invalid header length ($2 \times 4 = 8$).

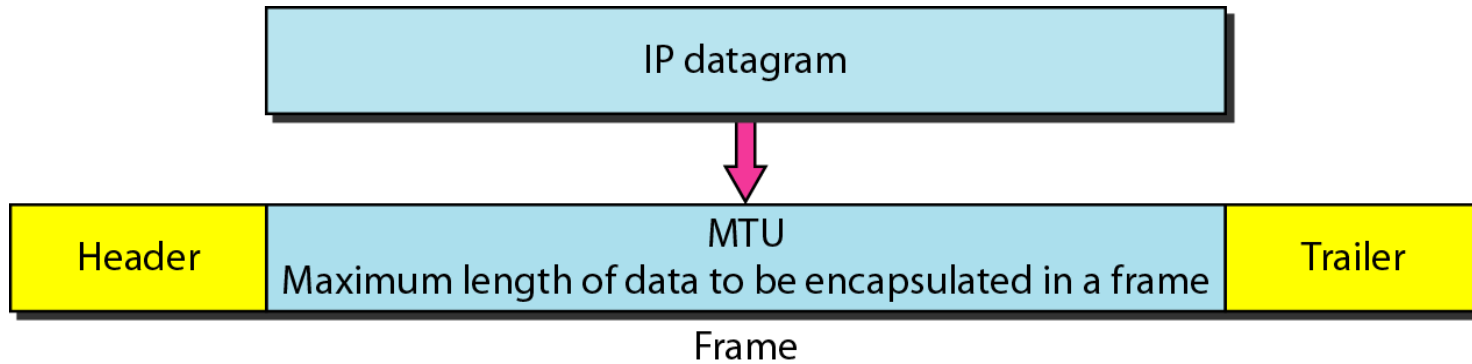
The minimum number of bytes in the header must be **20**. The packet has been corrupted in transmission.

Example: In an IPv4 packet, the value of HLEN is 5, and the value of the total length field is 0x0028. How many bytes of data are being carried by this packet?

Solution: The HLEN value is 5, which means the total number of bytes in the header is $5 \times 4 = 20$ bytes (no options).

The total length is 40 bytes, which implies that the packet is carrying **20** bytes of data ($40 - 20$).

Maximum transfer unit (MTU)



<i>Protocol</i>	<i>MTU</i>
Hyperchannel	65,535
Token Ring (16 Mbps)	17,914
Token Ring (4 Mbps)	4,464
FDDI	4,352
Ethernet	1,500
X.25	576
PPP	296

ADDRESS MAPPING

- A **packet** starting from a **source host** may **pass** through several **different physical networks** before finally reaching the **destination host**.
- The hosts and routers are recognized at the **network level** by their **logical (IP) addresses**.
- However, packets pass through physical networks to reach these hosts and routers,
- At the **physical level**, the hosts and routers are recognized by their **physical addresses**.

ADDRESS MAPPING

➤ We need to be able to map a logical address to its corresponding physical address and vice versa.

➤ This can be done by using either **static** or **dynamic mapping**.

In **dynamic mapping** each time a machine knows one of the two addresses (logical or physical), it can use a protocol to find the other one.

Mapping Logical to Physical

➤ Address Resolution Protocol (ARP)

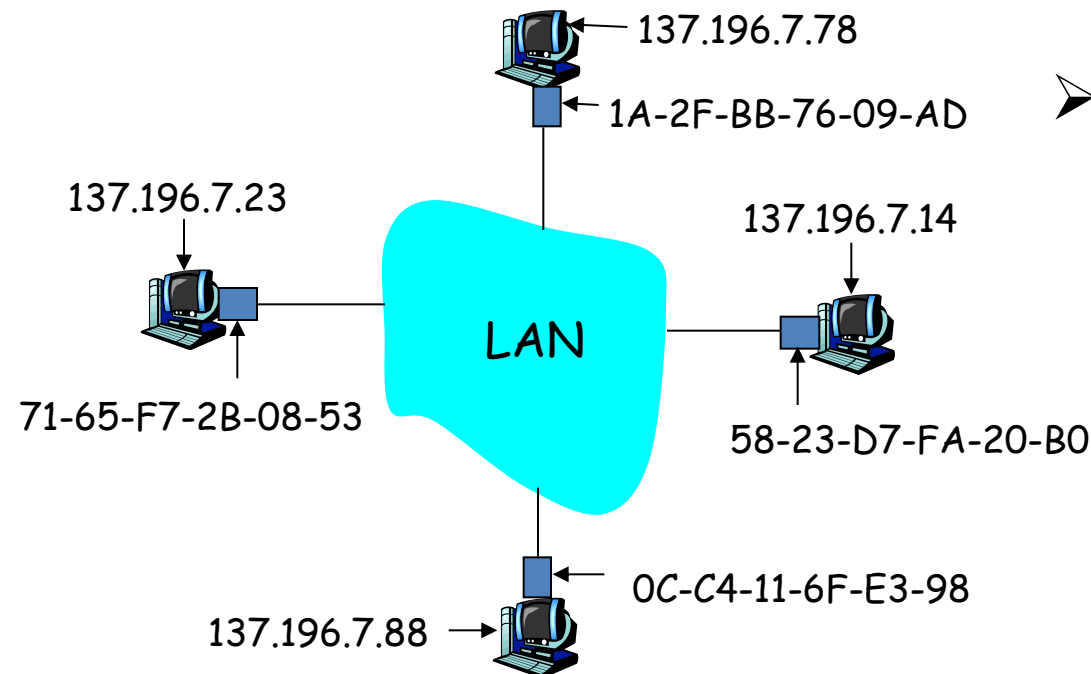
For Mapping Physical to Logical

➤ Reverse Address Resolution Protocol (RARP)

ARP: Address Resolution Protocol

Question: how to determine
MAC address of B
knowing B's IP address?

- Each IP node (Host, Router) on LAN has **ARP** table
- ARP Table: IP/MAC address mappings for some LAN nodes
- **< IP address; MAC address; TTL >**
- TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)



ARP protocol: Same LAN (network)

- A wants to send datagram to B, and B's MAC address not in A's ARP table.
- A **broadcasts** ARP query packet, containing B's IP address
 - Dest MAC address = FF-FF-FF-FF-FF-FF
 - all machines on LAN receive ARP query
- B receives ARP packet, replies to A with its (B's) MAC address
 - frame sent to A's MAC address (unicast)
- A caches (saves) IP-to-MAC address pair in its ARP table until information becomes old (times out)
 - soft state: information that times out (goes away) unless refreshed
- ARP is “plug-and-play”:
 - nodes create their ARP tables without intervention from net administrator

Internet Control Message Protocol (ICMP)

The IP protocol has no error-reporting or error-correcting mechanism.

The IP protocol also lacks a mechanism for host and management queries.

The ICMP has been designed to compensate for the above two deficiencies.

- ICMP always reports error messages to the original source

It is a companion to the IP protocol.

Internet Group Management Protocol (IGMP)

The IP protocol can be involved in two types of communication: unicasting and multicasting.

The IGMP is one of the necessary, but not sufficient, protocols that is involved in multicasting.

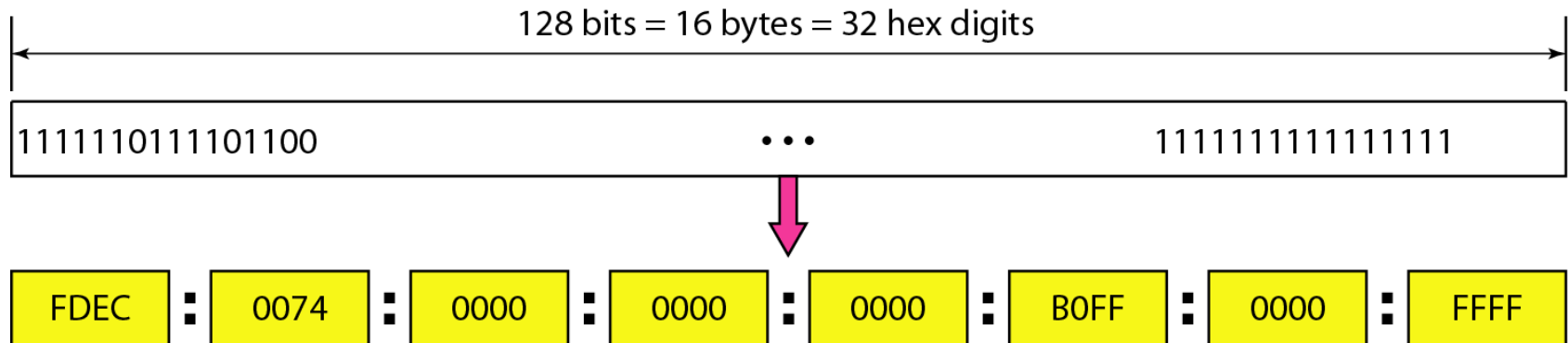
IGMP is a companion to the IP protocol.

IPv6 ADDRESSES

IPv6 ADDRESSES

Despite all short-term solutions, address depletion is still a long-term problem for the Internet. This and other problems in the IP protocol itself have been the motivation for IPv6.

An IPv6 address is 128 bits long



Abbreviated IPv6 addresses

Original

FDEC : 0074 : 0000 : 0000 : 0000 : B0FF : 0000 : FFF0



Abbreviated

FDEC : 74 : 0 : 0 : 0 : B0FF : 0 : FFF0



More abbreviated

FDEC : 74 : : B0FF : 0 : FFF0

Gap

EXAMPLE

Expand the address 0:15::1:12:1213 to its original.

Solution

We first need to align the left side of the double colon to the left of the original pattern and the right side of the double colon to the right of the original pattern to find how many 0s we need to replace the double colon.

XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX
0: 15: : 1: 12:1213

This means that the original address is.

0000:0015:0000:0000:0000:0001:0012:1213
